

EECS 489 Final Exam

Fall 2024

Name: _____

Uniqname: _____

INSTRUCTIONS

Welcome to the EECS 489 Fall 2024 Final.

- This exam is 120 minutes long and consists of both multiple choice and free response questions. Partial credit will be awarded where applicable, so please show all work and reasoning in your responses.
- This is a closed-book exam. You may use one note sheet, 8.5"×11", double-sided, with your name on it.
- Mark all solutions in the space provided. Any answers given outside the boxes will not be graded.
- Clearly fill your multiple choice answers. Erase completely if you change your answer. There is exactly one answer for every question.
- Bubble in your answer on the bar **above** the question. If your circled and bubbled answers are different, we will choose the answer that is bubbled in.
- There are questions on both sides of each page.
- The total points on the exam add up to 88.

Good luck!

When you are finished with the exam, sign below acknowledging the Honor Code

"I have neither given nor received unauthorized aid on this examination, nor have I concealed any violations of the Honor Code."

NAME:

THIS PAGE IS INTENTIONALLY LEFT BLANK

NAME: _____

Multiple Choice

(2 points each)

☐ A ☐ B ☐ C ☐ D

1. Suppose a link has a bandwidth of 20 Mbps. There are 4 flows with demands of 2 Mbps, 4 Mbps, 10 Mbps, and 6 Mbps over this link. What is the allocated bandwidth for each of the flows (in Mbps) using Min-Max fairness? Allocated bandwidth is listed in the same order as demands.

- a) 2, 4, 8, 6
- b) 2, 4, 9, 5
- c) 5, 5, 5, 5
- d) $\frac{20}{11}$, $\frac{40}{11}$, $\frac{100}{11}$, $\frac{60}{11}$

☐ A ☐ B ☐ C ☐ D

2. An IPv4 packet with a total size of 3000 bytes (including the IP header) is being sent from a device through two LANs. The first LAN has an MTU of 1500 bytes. What is the minimum number of fragments that are required to send this packet? You may assume that all IP headers are 20 bytes.

- a) 2
- b) 3
- c) 4
- d) 5

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

3. Suppose a router uses a **longest prefix match** algorithm for routing. If its routing table has the following entries:

Prefix	Netmask	Next Hop
192.168.0.0	255.255.0.0 (/16)	Router A
192.168.10.0	255.255.255.0 (/24)	Router B
192.168.10.128	255.255.255.128 (/25)	Router C
192.168.10.127	255.255.255.192 (/26)	Router D

Which next hop will an IP packet with a destination IP 192.168.10.130 go to?

- a) Router A
- b) Router B
- c) Router C
- d) Router D

Ⓐ Ⓑ Ⓒ Ⓓ

4. Which one of these is **not** an example of a mechanism that can be described as router-assisted congestion control?

- a) Explicit Congestion Notification (ECN)
- b) Weighted Fair Queuing
- c) Rate Control Protocol (RCP)
- d) Additive Increase, Multiplicative Decrease (AIMD)

Ⓐ Ⓑ Ⓒ Ⓓ

5. Which of the following [IPv4 header field]: [Purpose] pairings is **incorrect**?

- a) Time-to-live: Prevent routing loops
- b) Identification: Detect Corruption
- c) Fragment Offset: Handle Large Packets
- d) Protocol: Identify Packet Type

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

6. Which of the following statements regarding IPv6 is **incorrect**?

- a) IPv6 reduces the need for NAT (Network Address Translation)
- b) IPv6 uses the “Flow Label” field to enable fragmentation
- c) IPv6 lacks checksum functionality
- d) IPv6 has fixed-length headers

Ⓐ Ⓑ Ⓒ Ⓓ

7. When implementing the WTP-Base protocol in Project 3, Efe accidentally introduces a single bug. However, when he sanity-checks his code by transferring a large file, he manages to do so correctly. Which one of the following bugs is consistent with this behavior?

Recall that WTP-Base uses cumulative ACKs, with the receiver buffering out-of-order packets that are within the window. For each option, assume that no other bugs exist. Assume the file is large enough that it requires at least twice the `window_size` number of packets to transmit. Assume the link has non-zero propagation delay, transmission delay, and packet loss rate.

- a) `wReceiver` sends selective ACKs, sending an ACK with sequence number i in response to a packet with sequence number i .
- b) When `wReceiver` calls `recvfrom` using a UDP socket to get a new packet, it supplies a buffer of size 1 in order to read byte-by-byte.
- c) `wSender` numbers packets starting at 1 instead of 0.
- d) `wSender` advances the sender-side window only when an ACK arrives that indicates the last packet in the window was received.

Ⓐ Ⓑ Ⓒ Ⓓ

8. Aditya is now implementing the WTP-OPT protocol for Project 3; he also accidentally introduces a single bug. He notices that his code successfully manages to transfer files sometimes, but not always. Which one of the following bugs is consistent with this behavior?

Recall that WTP-OPT uses selective ACKs. For each option, assume that no other bugs exist. Assume the file requires at least twice the `window_size` number of packets to transmit. Assume the link has non-zero propagation delay, transmission delay, and packet loss rate.

- a) A `sleep` call in `wReceiver` causes ACKs to be sent after 1 second of receiving the packet, despite the sender-side timeout being 500ms.
- b) `wReceiver` sends cumulative ACKs, sending an ACK with sequence number equal to the next expected sequence number.
- c) `wSender` resends all packets in the window instead of just the unacknowledged packets when any packet times out.
- d) `wSender` shifts the window by one packet every time an ACK is received, ignoring the sequence number of the ACK.

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

9. Which of the following is an accurate statement about software-defined networking (SDN)?

- a) SDN involves a centralized control plane and distributed data plane.
- b) An SDN controller routes incoming packets to different routers for load-balancing.
- c) Routers using SDN are slower than traditional routers as a remote SDN controller must be consulted to route every packet rather than a local routing table.
- d) SDN is only applicable in data centers and enterprise networks.

Ⓐ Ⓑ Ⓒ Ⓓ

10. Which of the following correctly describes a difference between *infrastructure mode* and *ad hoc mode* in a wireless network?

- a) Infrastructure mode requires a central access point, while ad hoc mode allows devices to communicate directly without a central access point.
- b) Infrastructure mode creates point-to-point communication, while ad hoc mode enables broadcast communication.
- c) Ad hoc mode is more scalable than infrastructure mode as there is less overhead created from infrastructure concerns.
- d) Wireless access points in infrastructure mode are connected by wire to L2 switches; on the other hand, wireless access points in ad hoc mode are connected wirelessly to L3 routers.

Ⓐ Ⓑ Ⓒ Ⓓ

11. Which of the following is not an application of tunneling?

- a) Creating a Virtual Private Network (VPN)
- b) Supporting new protocols, such as IPv6
- c) Optimizing network traffic by forcing packets to use specific paths
- d) Transmitting large packets that do not fit in a single Ethernet frame

Ⓐ Ⓑ Ⓒ Ⓓ

12. Which of the following is **not** a disadvantage of Network Address Translation (NAT)?

- a) NAT adds additional complexity for machines behind different NAT devices to establish direct connections, as each device no longer has a unique IP address.
- b) NAT reduces security, as packets can no longer be encrypted in order for the NAT device to translate IP addresses.
- c) NAT adds some translation overhead, potentially increasing latency and consuming processing power.
- d) NAT can complicate network troubleshooting by hiding IP addresses.

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

13. What is a disadvantage of using topology-aware addressing in data center networks?

- a) It reduces network scalability as the addressing scheme grows too complex.
- b) It complicates routing decisions and increases overhead by necessitating complex forwarding tables that align with network topology.
- c) It leads to inefficient use of available bandwidth and may cause congestion in certain parts of the network.
- d) It requires regular updates to the addressing scheme to reflect the dynamic changes in the network topology.

Ⓐ Ⓑ Ⓒ Ⓓ

14. Suppose that you are a router in a network that uses distance vector updates for calculating the shortest path to a destination with poisoned reverse. Suppose that your currently calculated distance vector to other routers is

$$\begin{pmatrix} k & u & v & z \\ 5 & 7 & 2 & 10 \end{pmatrix}$$

Router k then sends you the following updated distance vector message:

$$\begin{pmatrix} k & u & v & z & m \\ 0 & 3 & 5 & 4 & 7 \end{pmatrix}$$

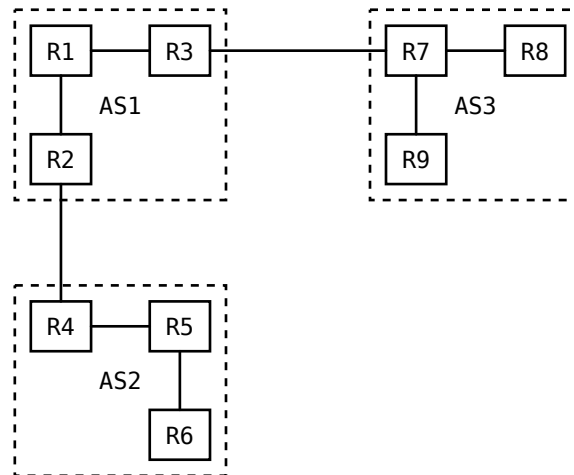
Which of the following is the correctly updated distance vector that belongs to you, including any routers that were previously unknown to us?

- a) $\begin{pmatrix} k & u & v & z & m \\ 5 & 7 & 2 & 10 & 7 \end{pmatrix}$
- b) $\begin{pmatrix} k & u & v & z & m \\ 5 & 7 & 2 & 9 & 12 \end{pmatrix}$
- c) $\begin{pmatrix} k & u & v & z & m \\ 5 & 3 & 2 & 10 & 7 \end{pmatrix}$
- d) $\begin{pmatrix} k & u & v & z & m \\ 5 & 8 & 10 & 9 & 12 \end{pmatrix}$

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

15. For this question, refer to the below network diagram. Each dashed square represents an AS. The ASes and the routers within use BGP to communicate about network paths. Assume that all ASes in the diagram have peer-peer relationships and use standard Gao-Rexford rules to decide which paths to communicate.



How do R1, R4, R6, and R8 learn about a subnet accessible only through R3?

Option	R1	R4	R6	R8
I	IGP	eBGP	eBGP	iBGP
II	iBGP	eBGP	iBGP	iBGP
III	iBGP	eBGP	IGP	IGP
IV	IGP	eBGP	iBGP	iBGP

- a) I
- b) II
- c) III
- d) IV

Ⓐ Ⓑ Ⓒ Ⓓ

16. Which of the following is **not** an accurate statement regarding Intra-AS routing?

- a) Link-state routing tends to have smaller messages than distance-vector routing.
- b) Distance-vector routing can take much longer to converge than link-state routing.
- c) Distance-vector routing is more similar to BGP routing than link-state routing.
- d) The poisoned reverse heuristic can help both distance-vector and link-state routing eliminate count-to-infinity scenarios.

NAME: _____

Ⓐ Ⓑ Ⓒ Ⓓ

17. Which of the following is **not** an accurate statement regarding IP and MAC addresses?

- a) MAC addresses are associated with individual network interfaces rather than devices.
- b) There are more IPv6 addresses than MAC addresses.
- c) IPv4 addresses are always permanently assigned, but can easily be spoofed.
- d) An ARP request is intended to discover the MAC address corresponding to an IP address.

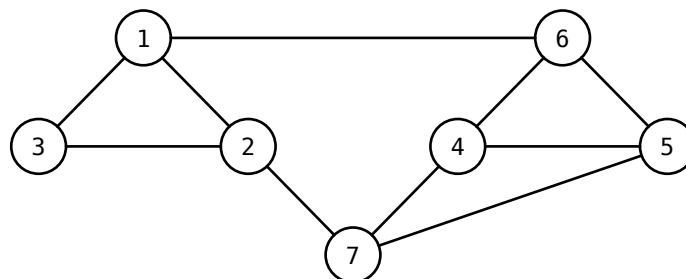
Ⓐ Ⓑ Ⓒ Ⓓ

18. Which of the following is the Dynamic Host Configuration Protocol (DHCP) used for?

- a) Discovering the subnet mask of the current network
- b) Discovering the IP address that maps to a certain URL
- c) Sending error messages when packets are unable to be delivered
- d) Creating a spanning tree to prevent broadcast storms on a switched Ethernet network

Ⓐ Ⓑ Ⓒ Ⓓ

19. Consider the following graph of nodes in a switched Ethernet network. Suppose these nodes construct a spanning tree amongst themselves using the algorithm learned in class, electing the node with the smallest identifier as the root. Which of the following edges will **not** be part of the spanning tree?



- a) (4, 7)
- b) (5, 6)
- c) (1, 3)
- d) (4, 6)

Short Answer

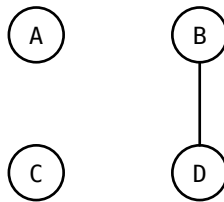
20. Poisoned Reverse

(10 points)

20.1 (*4 points*) Suppose a distance-vector protocol is being used for intra-AS routing. Please draw any additional edges between the nodes as you see fit and assign costs to all edges such that deleting the edge BD creates

- a count-to-infinity scenario between A and D if poisoned reverse is not used.
- no issue if poisoned reverse is used.

Briefly explain your answer.

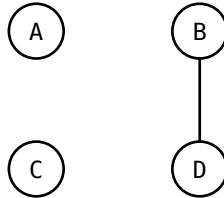


Answer:

NAME: _____

20.2 (4 points) Suppose a distance-vector protocol is being used **with** poisoned reverse. Please draw any additional edges between the nodes as you see fit and assign costs to all edges such that deleting the edge BD still creates a count-to-infinity scenario between A and D .

Briefly explain your answer.



Answer:

20.3 (2 points) Does BGP routing encounter count-to-infinity scenarios? Why or why not? Explain in 2-3 sentences.

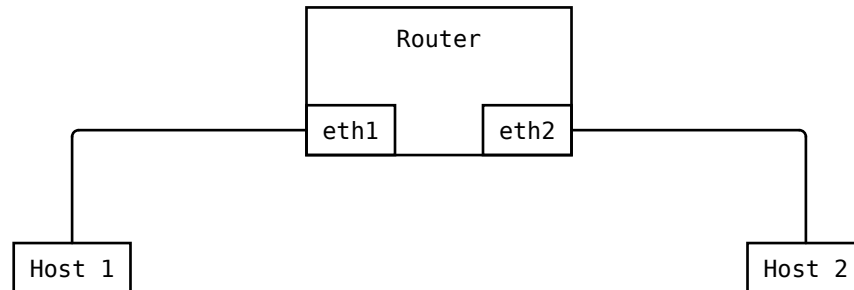
Answer:

NAME: _____

21. Sending a Packet

(15 points)

Consider the following network topology:



Host 1 and Host 2 are on different local area networks (LANs), with the router connecting the two hosts. The router and the hosts each have their own ARP Cache to keep track of IP-to-MAC mappings; all ARP Caches are initially empty. This is very similar to the setup in Project 4.

Some information about the network interfaces being used are given below:

Interface	MAC Address	IP Address
Host 1	A	B
Host 2	C	D
eth 1	E	F
eth 2	G	H

NAME: _____

21.1 (5 points) Host 1 would like to send a UDP packet to Host 2. Please indicate all packets that will arrive and leave the router in chronological order in the following table. You may not need to use all rows.

In/Out of Router	Packet Type (ARP Request/ARP Reply/ICMP/UDP)	Router Interface (eth1/eth2)

21.2 (5 points) Fill in the blank fields for the indicated layers on the UDP packet that Host 1 would send to communicate with Host 2 in 21.1.

Ethernet Layer	
Source MAC	
Destination MAC	

IP Layer	
Source IP	
Destination IP	

NAME: _____

21.3 (5 points) Now, Host 1 decides to send a ping (echo) request to Host 2. Once Host 2 receives this ping request, it will reply. Please indicate all packets that will enter and leave the router as a result of this exchange (including the reply) in chronological order.

Assume that any ARP entries cached from the previous exchange are still valid, and no other ARP entries have been created. As in Project 4, assume that replying to echo requests does not require IP-to-MAC resolution. Once again, you may not need to use all rows.

In/Out of Router	Packet Type (ARP Request/ARP Reply/ICMP/UDP)	Router Interface (eth1 / eth2)

NAME:

THIS PAGE IS INTENTIONALLY LEFT BLANK

NAME: _____

22. “Switching” Up P4

In Project 4, we implemented a router that learned the MAC addresses of different routers using ARP requests. In lecture, we discussed how bridges on local networks instead use self-learning to know how to route incoming packets on a local network. In this question, you will be implementing a self-learning bridge.

Refer to the reference sheet on the following page for the structure of the class.

You will be implementing the `LearningBridge::handlePacket` function. You may assume that all other functions have been implemented and function correctly. For simplicity, assume that links are connected in a spanning-tree configuration, so you do not need to worry about broadcast storms.

```
void LearningBridge::handlePacket(EthernetPacket packet, std::string port) {
```


Reference for Question 22

```
struct EthernetPacket {
    mac_addr source_mac;
    mac_addr dest_mac;
    std::vector<uint8_t> data;
};

/**
 * @class LearningBridge A switch that learns the ports different hosts are located on
 * by inspecting incoming packets.
 */
class LearningBridge {
public:
    /**
     * @brief Called when an ethernet packet is received on a port. This function ensures
     * that the switch "learns" from the incoming packet if possible.
     *
     * If the destination is not associated with a port, the packet should be broadcasted.
     * If the destination is associated with a port, the packet should only be sent out
     * on that port.
     */
    void handlePacket(EthernetPacket packet, std::string port);
private:
    /**
     * @brief Sends a packet out on the specified port
     */
    void sendPacket(EthernetPacket packet, std::string port);
    /**
     * @brief Associates a host with the MAC address specified with the specified port
     */
    void associatePortWithMac(mac_addr mac, std::string port);
    /**
     * @brief Return the port associated with the host with the given MAC address,
     * if such an association exists
     */
    std::optional<std::string> getPortForMac(mac_addr mac);

    /**
     * @brief Return a vector of all the ports that exist on the switch
     */
    const std::vector<std::string>& getAllPorts();

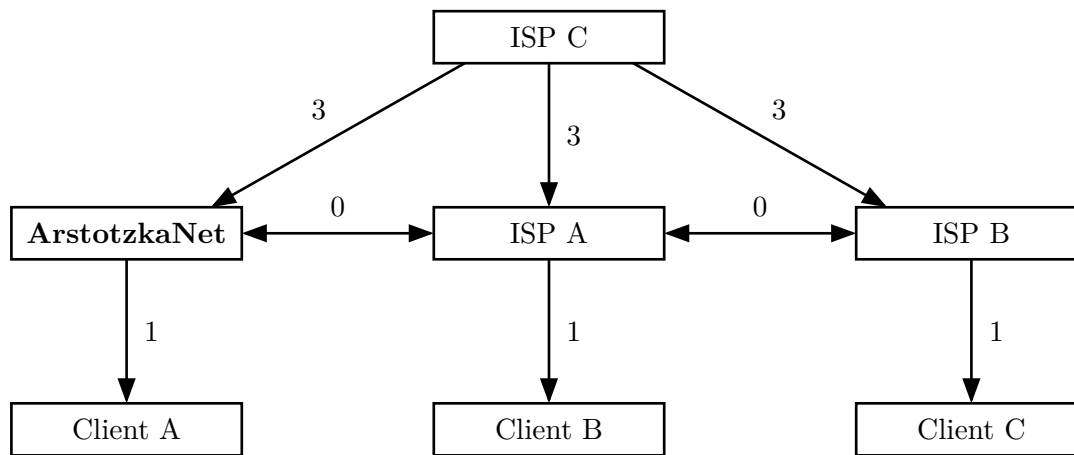
    // Member data structures not shown
};
```

23. BGP, Please

(10 points)

Congratulations! You have been promoted to a position where you are in charge of BGP configurations for a global internet service provider (ISP) called ArstotzkaNet. You are responsible for configuring and enforcing BGP agreements such that all of the ISP's routers only route packets through autonomous systems (ASes) that they are allow to use for routing. You should assume that Gao-Rexford rules are being followed.

The BGP graph for your ISP is given in the following diagram. The edge weights represent the cost of routing packets through each connection.



$A \leftrightarrow B$: A and B have a peering connection
 $A \rightarrow B$: A is a provider of B

23.1 (3 points) Name the cheapest path that Client A can use to communicate with Client B. If there is a cheaper path that Client A cannot use because of BGP rules, write down the path and the reason it cannot be used.

Answer:

NAME:

23.2 (3 points) Name the cheapest path that Client A can use to communicate with Client C. If there is a cheaper path that Client A cannot use because of BGP rules, write down the path and the reason it cannot be used.

Answer:

23.3 (4 points) Suppose that the government in your country has ordered that access to IP addresses within the `umich.edu` AS be blocked. Assume the `umich.edu` AS is a customer of ISP C that is not shown in this diagram.

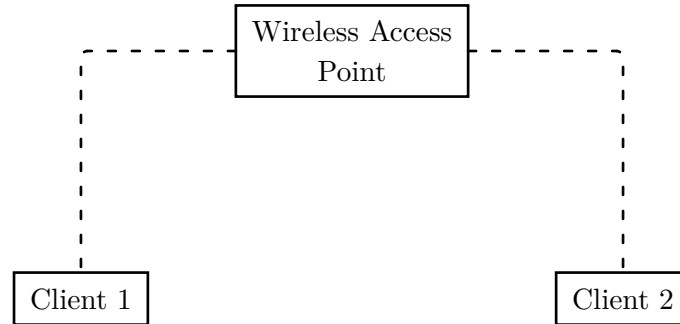
Explain how ArstotzkaNet can take advantage of BGP to cause both Client A and ISP A to not be able to access the `umich.edu` AS (at least temporarily). For your answer to this question, ArstotzkaNet need not follow Gao-Rexford rules.

Answer:

24. Look Ma, No Wires!

(15 points)

Consider the below 802.11 wireless network setup below.



Assume that:

- The bandwidth of each wireless link is 1 MB/s.
- The one-way propagation delay between either client and the wireless access point is 0.5 seconds.
- The clients are not in direct range of each other.
- A client has a timeout of 1 RTT (i.e. 1s) for backing off after sending a packet if the expected response is not received.
- No packets are dropped or lost.

Both Client 1 and Client 2 select backoff times for CSMA/CA according to a function $B(i, k)$ such that

$$B(i, k) = 2(k + 1) \cdot i$$

where i is the number of the client, and k is the number of times the client has backed off previously.

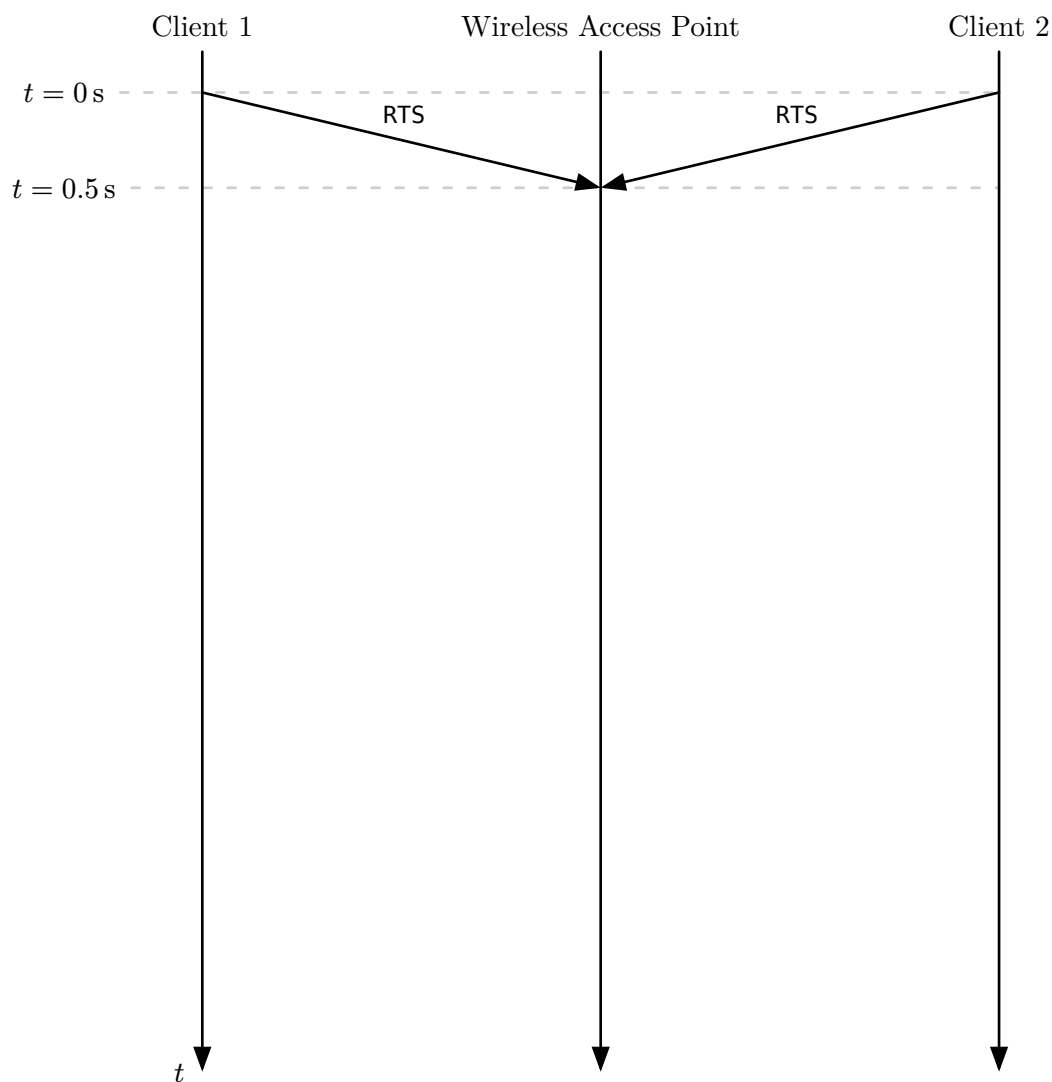
For example, if Client 1 backs off for the first time, $i = 1$ and $k = 0$.

NAME: _____

24.1 (9 points) Both Client 1 and Client 2 would both like to send 5 MB of data to the wireless access point. Both clients attempt to start transmitting at $t = 0$.

Below, draw the order in which different collision avoidance messages and actual data are sent until Client 1 completes its transmission, including receiving the acknowledgement message.

- As usual, time is on the vertical axis. The vertical axis does not need to be to-scale.
- Make sure to clearly indicate the type of message, the time transmission begins, the time the message is fully received, and the direction of the message.
- Keep in mind that wireless is a broadcast medium.

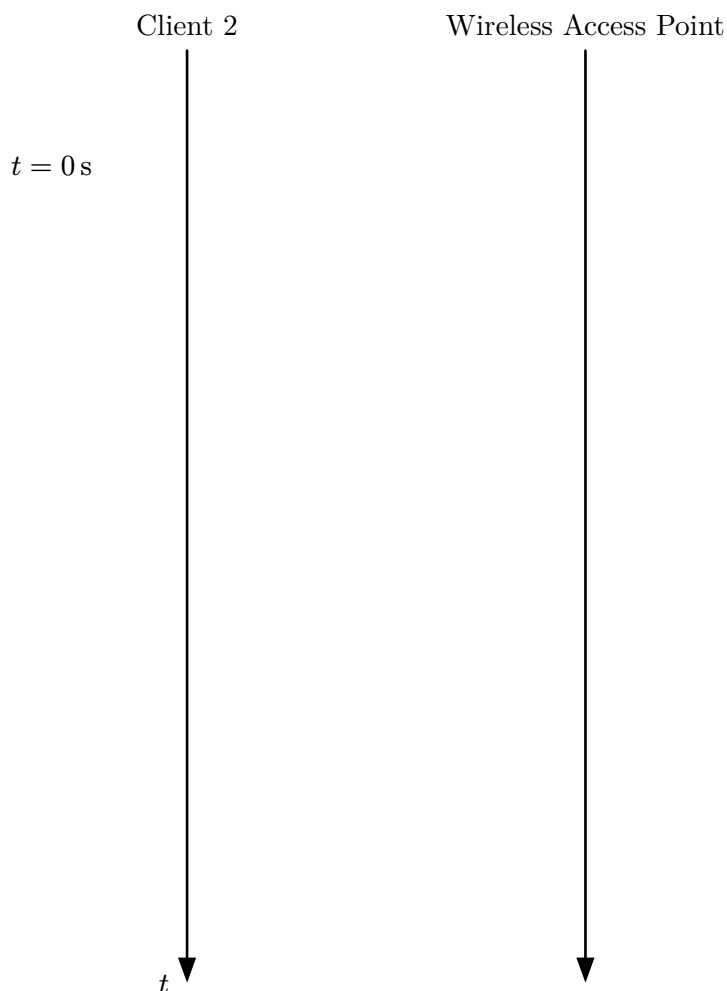


NAME: _____

24.2 (6 points) After Client 1 finishes its transmission, Client 2 attempts to transmit its 5 MB of data again.

Below, draw the order in which different collision avoidance messages and actual data are sent until Client 2 completes its transmission, including receiving its acknowledgement message.

- You should use $t = 0$ as the start time for Client 2's transmission (i.e. do **not** continue from the final time on question 24.1).
- As usual, time is on the vertical axis. The vertical axis does not need to be to-scale.
- Make sure to clearly indicate the type of message, the time transmission begins, the time the message is fully received, and the direction of the message.
- Keep in mind that wireless is a broadcast medium.



NAME:

THIS PAGE IS INTENTIONALLY LEFT BLANK

NAME:

THIS PAGE IS INTENTIONALLY LEFT BLANK